**Purdue University** **2026-2027 University Catalog**

# Civil Engineering, BSCE

**Location(s):** West Lafayette

## About the Program

The BS program in Civil Engineering is accredited by the Engineering Accreditation Commission of ABET, under the commission's General Criteria and Program Criteria for Civil and Similarly Named Engineering Programs.

Purdue civil engineers are shaping the world! From the communities we live in, to the air we breathe and water we drink, civil engineers are constantly working to make the planet a better place to live.

Civil engineering is a diverse and incredibly wide-reaching field that affects nearly every facet of our lives. Civil engineers are the creators of our built environment. They design our skyscrapers and stadiums, expand our harbors and ports, provide us access to energy, and ensure safe travel across our bridges and through our airports.

Civil engineers are also stewards of our natural environment. They study weather shifts, ocean coasts, and wetlands to ensure our ecosystems are sustained. They work to improve air quality and ensure everyone has access to clean water.

Innovation is synonymous with civil engineering. As the world becomes more digital, so have we. Civil engineers are designing smart, sustainable, self-regulating buildings; they are harnessing heat and vibration by converting them to electricity; and are designing roads that will charge electric vehicles.

You can explore the nine areas of study within civil engineering along with selecting the courses to design your own plan of study. You and your advisor can discuss your career goals to tailor a program to meet your goals.

Instructional laboratories in structural behavior, hydraulics, geomatics, and civil engineering materials are offered in the sophomore and junior years. Further study includes 30 credits of technical electives allowing students to tailor their studies to their specialty area of choice. Specialty areas include architectural, construction, environmental, geomatics, geotechnical, hydraulics and hydrology, materials, structures, and transportation and infrastructure systems engineering.

A Senior Design Capstone course culminates undergraduate students' academic careers. This course allows students to connect with real-world projects and develop and build skills such as problem solving, engineering design, teamwork, communication, project management, and project planning. Student teams act as their own engineering companies and present their work throughout the semester via presentations and written reports.

Experiential learning opportunities within Civil Engineering are encouraged and include: Study Abroad experiences either short-term or semester long programs, Undergraduate research and service learning (EPICS), Internships and Co-ops, clubs and organizations, and the honors program.

A combined BSCE + MSCE program is available for outstanding Civil Engineering undergraduate students. This program can be completed in as little as five years (non-thesis and thesis option MSCE are possible) and result in receiving both the BSCE and MSCE degrees.

Lyles School of Civil and Construction Engineering

Civil Engineering Major Change (CODO) Requirements

## Degree Requirements

### 130 Credits Required

## Civil Engineering Major Requirements (65 credits)

### Civil Engineering Required Major Courses (35 credits)

A GPA of 2.0 and a grade of C- or better is required for all CCE courses in Required Major Courses section. (CE Core Course Policy)

- CCE 20300 - Principles And Practice Of Geomatics **Credit Hours: 4.00**
- CCE 21100 - Thermal And Energy Sciences **Credit Hours: 3.00**
- CCE 27000 - Introductory Structural Mechanics **Credit Hours: 4.00** ♦
- CCE 29200 - Contemporary Issues In Civil Engineering **Credit Hours: 2.00**
- CCE 29700 - Basic Mechanics I (Statics) **Credit Hours: 3.00** ♦
- CCE 29800 - Basic Mechanics II Dynamics **Credit Hours: 3.00** ♦
- CCE 33500 - Civil Engineering Materials **Credit Hours: 4.00**
- CCE 34000 - Hydraulics **Credit Hours: 3.00**
- CCE 34300 - Elementary Hydraulics Laboratory **Credit Hours: 1.00**
- CCE 39800 - Introduction To Civil Engineering Systems Design **Credit Hours: 3.00**
- CCE 39200 - Technical Communication In Civil Engineering **Credit Hours: 2.00**
- CCE 49800 - Civil Engineering Design Project **Credit Hours: 3.00**

### Technical Selectives (30 credits)

- Detailed requirements for Technical Selectives are outlined in the Civil Engineering Supplemental Information link below.
- Technical Selective courses include CCE 30000, 40000, and 50000 level courses that are not used to fulfill Required Major Course requirements.

# Other Departmental/Program Course Requirements (65-75 credits)

## Other Course Requirements (44-51 credits)

Grade of C- of better is required.

**Intro to Engineering I & II** - Credit Hours:4.00-8.00

- ENGR 13100 - Transforming Ideas To Innovation I (UCC: AI/IL) **Credit Hours: 2.00** and
- ENGR 13200 - Transforming Ideas To Innovation II (UCC: AI) **Credit Hours: 2.00**

OR

- EPCS 11100 - First Year Participation In EPICS I (UCC: STS) **Credit Hours: 1.00** and
- EPCS 12100 - First Year Participation In EPICS II (UCC: STS) **Credit Hours: 1.00** and
- ENGR 13300 - Transforming Ideas To Innovation, EPICS/VIP (UCC: AI/IL) **Credit Hours: 2.00**

OR

- VIP 17911 - First Year Participation In Vertically Integrated Projects (VIP) I (UCC: STS) **Credit Hours: 1.00** and
- VIP 17912 - First Year Participation In Vertically Integrated Projects (VIP) II **Credit Hours: 1.00** and
- ENGR 13300 - Transforming Ideas To Innovation, EPICS/VIP (UCC: AI/IL) - **2.00** **Credit Hours**

OR

- ENGR 16100 - Honors Introduction To Innovation And The Physical Science Of Engineering Design I (UCC: AI/IL) **Credit Hours: 4.00** and
- ENGR 16200 - Honors Introduction To Innovation And The Physical Science Of Engineering Design II (UCC: SCI) **Credit Hours: 4.00**

OR

- ENGR 13000 - Transforming Ideas Into Innovations (UCC: AI/IL) **Credit Hours: 4.00**

- CHM 11510 - General Chemistry I **Credit Hours: 3.00**

AND

- CHM 11520 - General Chemistry I - Laboratory (UCC: SCI) **Credit Hours: 1.00** or
- CHM 11530 - General Chemistry I - Virtual Laboratory (UCC: SCI) **Credit Hours: 1.00**

- CS 15900 - C Programming **Credit Hours: 3.00**

OR

- CHM 11610 - General Chemistry II **Credit Hours: 3.00**

AND

- ( CHM 11620 - General Chemistry II - Laboratory (UCC: SCI) **Credit Hours: 1.00** or
- CHM 11630 - General Chemistry II - Virtual Laboratory (UCC: SCI) **Credit Hours: 1.00** )

- MA 16100 - Plane Analytic Geometry And Calculus I (UCC: QR) **Credit Hours: 5.00** or
- MA 16500 - Analytic Geometry And Calculus I (UCC: QR) **Credit Hours: 4.00**

- MA 16200 - Plane Analytic Geometry And Calculus II (UCC: QR) **Credit Hours: 5.00** or
- MA 16600 - Analytic Geometry And Calculus II (UCC: QR) **Credit Hours: 4.00**

**Physics I** - Credit Hours: 4.00

- PHYS 17200 - Modern Mechanics (UCC: SCI) **Credit Hours: 4.00**

OR

- ENGR 16100 - Honors Introduction To Innovation And The Physical Science Of Engineering Design I (UCC: AI/IL) - **4.00** **Credit Hours** and
- ENGR 16200 - Honors Introduction To Innovation And The Physical Science Of Engineering Design II (UCC: SCI) - **4.00** **Credit Hours**

- CM 16400 - Construction Graphics And Digital Documentation **Credit Hours: 2.00** ♦
- MA 26100 - Multivariate Calculus (UCC: QR) **Credit Hours: 4.00** ♦
- MA 26500 - Linear Algebra (UCC: QR) **Credit Hours: 3.00**
- MA 26600 - Ordinary Differential Equations (UCC: QR) **Credit Hours: 3.00**
- PHYS 24100 - Electricity And Optics (UCC: SCI) **Credit Hours: 3.00** ♦
- STAT 51100 - Statistical Methods **Credit Hours: 3.00**
- Basic Science Selective - Credit Hours: 3.00 (See Supplemental information for list of courses)

## Engineering General Education Requirements (21 credits)

For a list of courses that satisfy this requirement, see Supplemental Information.

- General Education I (UCC: HUM) - Credit Hours: 3.00-4.00
- General Education II (UCC: BSS) - Credit Hours: 3.00
- General Education III - Credit Hours: 3.00
- General Education IV - Credit Hours: 3.00
- General Education V - Credit Hours: 3.00
- Written Communication (UCC: WC) - Credit Hours: 3.00-4.00
- Oral Communication (UCC: OC) - Credit Hours: 3.00

### Additional General Education Requirements

- At least 6 of these 21 credits must be in non-introductory courses (30000-level or above OR from courses with a required pre-requisite in the same department)
- At least 12 of these 21 credits must be taken from the College of Liberal Arts, the Daniels School of Business, and/or the Honors College provided such courses are not focused primarily on engineering, technology, the natural sciences, or mathematics.

- A course is not eligible for use towards these 21 credits of Engineering General Education requirement if that course is used to satisfy a different engineering degree requirement.

## Supplemental Information

- Click here for Civil Engineering Supplemental Information
- Click here for Civil Engineering General Education Courses

## Optional Concentrations

- Architectural Engineering Concentration for Civil Engineering, BSCE
- Construction Engineering Concentration for Civil Engineering
- Environmental Engineering Concentration for Civil Engineering
- Geomatics Engineering Concentration in Civil Engineering
- Geotechnical Engineering Concentration in Civil Engineering
- Hydraulic and Hydrologic Engineering Concentration in Civil Engineering
- Materials Engineering Concentration in Civil Engineering
- Structural Engineering Concentration for Civil Engineering
- Transportation and Infrastructure Systems Engineering Concentration for Civil Engineering

## GPA Requirements

- A GPA of 2.0 is required for all Baccalaureate degrees.
- Students must have a 2.0 GPA in all “CCE” designated courses.

## Grade Requirements

- Students must have a grade of C- or better in all courses except Technical Selectives and General Education courses. See the CE Core Course Policy for more information.

## Pass/No Pass Policy

- Courses with a Pass/No Pass grade (P/NP) may not be used to satisfy degree requirements.

## Notes

- Students in Civil Engineering, BS cannot also enroll in the Construction Engineering, BS.

## Transfer Credit Policy

If you are interested in registering for a course offered by a different institution, please review the information on the Purdue Office of the Registrar Transfer Credit website.
* If you are an admitted transfer student, please review the above Transfer Credit website and then work with your academic advisor to determine exactly how your previous courses might apply.

## University Requirements

### University Core Requirements

**FOR A COMPLETE LISTING OF UNIVERSITY CORE COURSE SELECTIVES, VISIT THE UNIVERSITY SENATE WEBSITE.**

- AI Working Competency (UCC: AI)
- Human Cultures: Behavioral/Social Science (UCC: BSS)
- Human Cultures: Humanities (UCC: HUM)
- Information Literacy (UCC: IL)
- Oral Communication (UCC: OC)
- Quantitative Reasoning (UCC: QR)
- Science #1 (UCC: SCI)
- Science #2 (UCC: SCI)
- Science, Technology, and Society (UCC: STS)
- Written Communication (UCC: WC)

### Civics Literacy Proficiency Requirement

The Civics Literacy initiative is designed to develop civic knowledge of Purdue students to graduate a more informed citizenry. For more information visit the Civics Literacy - Office of the Vice Provost for Teaching and Learning - Purdue University.

To obtain the Civics Literacy credential, all Baccalaureate degree seeking students will complete the Civics Literacy Podcast pathway - Complete 12 podcasts created by the Purdue Center for C-SPAN Scholarship & Engagement that use C-SPAN material and pass the required quizzes per episode. After the pathway has been successfully completed, students will take the Civics Literacy knowledge test. Civics Literacy does not impact a student's GPA or bear any financial costs.

More details about each pathway and how to complete the requirement can be found on the Civics Literacy Student Dashboard in myPurdue.

### Upper Level Requirement

- Resident study at Purdue University for at least two semesters and the enrollment in and completion of at least 32 semester hours of coursework required and approved for the completion of the degree. These courses are expected to be at least junior-level (30000+) courses.
- Students should be able to fulfill *most*, *if not all*, of these credits within their major requirements; there should be a clear pathway for students to complete any credits not completed within their major.

## Sample Plan of Study

The following sample plan of study is a suggested arrangement of courses. Consultation with an advisor may result in an altered plan customized for an individual student.

### Fall 1st Year

- Intro to Engineering I – Credit Hours: 2.00-4.00
- Written Communication (UCC: WC) or Oral Communication (UCC:OC) - Credit Hours: 3.00-4.00
- CHM 11510 - General Chemistry I **Credit Hours: 3.00**

- CHM 11520 - General Chemistry I - Laboratory (UCC: SCI) **Credit Hours: 1.00** or
- CHM 11530 - General Chemistry I - Virtual Laboratory (UCC: SCI) **Credit Hours: 1.00**

- MA 16100 - Plane Analytic Geometry And Calculus I (UCC: QR) **Credit Hours: 5.00** or
- MA 16500 - Analytic Geometry And Calculus I (UCC: QR) **Credit Hours: 4.00**

#### 13-17 Credits

### Spring 1st Year

- Intro to Engineering II - Credit Hours: 2.00-4.00
- Written Communication (UCC: WC) or Oral Communication (UCC:OC) - Credit Hours: 3.00-4.00
- Physics I - Credit Hours: 4.00

- CS 15900 - C Programming **Credit Hours: 3.00**
  OR
- CHM 11610 - General Chemistry II **Credit Hours: 3.00**
  AND
- ( CHM 11620 - General Chemistry II - Laboratory (UCC: SCI) **Credit Hours: 1.00** or
- CHM 11630 - General Chemistry II - Virtual Laboratory (UCC: SCI) **Credit Hours: 1.00** )

- MA 16200 - Plane Analytic Geometry And Calculus II (UCC: QR) **Credit Hours: 5.00** or
- MA 16600 - Analytic Geometry And Calculus II (UCC: QR) **Credit Hours: 4.00**

**16-21 Credits**

## Fall 2nd Year

- MA 26100 - Multivariate Calculus (UCC: QR) **Credit Hours: 4.00** ♦
- PHYS 24100 - Electricity And Optics (UCC: SCI) **Credit Hours: 3.00** ♦
- CCE 29700 - Basic Mechanics I (Statics) **Credit Hours: 3.00** ♦
- CCE 20300 - Principles And Practice Of Geomatics **Credit Hours: 4.00**
- CM 16400 - Construction Graphics And Digital Documentation **Credit Hours: 2.00** ♦
- CCE 29200 - Contemporary Issues In Civil Engineering **Credit Hours: 2.00**

**18 Credits**

## Spring 2nd Year

- MA 26500 - Linear Algebra (UCC: QR) **Credit Hours: 3.00**
- CCE 21100 - Thermal And Energy Sciences **Credit Hours: 3.00**
- CCE 27000 - Introductory Structural Mechanics **Credit Hours: 4.00**
- CCE 29800 - Basic Mechanics II Dynamics **Credit Hours: 3.00**
- General Education I (UCC: HUM) - Credit Hours: 3.00-4.00

**16 Credits**

## Fall 3rd Year

- MA 26600 - Ordinary Differential Equations (UCC: QR) **Credit Hours: 3.00**
- CCE 33500 - Civil Engineering Materials **Credit Hours: 4.00**
- CCE 34000 - Hydraulics **Credit Hours: 3.00**
- CCE 34300 - Elementary Hydraulics Laboratory **Credit Hours: 1.00**
- Technical Selective I - Credit Hours: 3.00
- General Education II (UCC: BSS) - Credit Hours: 3.00

### 17 Credits

## Spring 3rd Year

- STAT 51100 - Statistical Methods **Credit Hours: 3.00**
- CCE 39800 - Introduction To Civil Engineering Systems Design **Credit Hours: 3.00**
- CCE 39200 - Technical Communication In Civil Engineering **Credit Hours: 2.00**
- Technical Selective II - Credit Hours: 3.00
- Technical Selective III - Credit Hours: 3.00
- Technical Selective IV - Credit Hours: 3.00

### 17 Credits

## Fall 4th Year

- Basic Science Selective - Credit Hours: 3.00
- Technical Selective V - Credit Hours: 3.00
- Technical Selective VI - Credit Hours: 3.00
- Technical Selective VII - Credit Hours: 3.00
- General Education III - Credit Hours: 3.00
- General Education IV - Credit Hours: 3.00

### 18 Credits

## Spring 4th Year

- CCE 49800 - Civil Engineering Design Project **Credit Hours: 3.00**
- Technical Selective VIII - Credit Hours: 3.00
- Technical Selective IX - Credit Hours: 3.00
- Technical Selective X - Credit Hours: 3.00
- General Education V - Credit Hours: 3.00

### 15 Credits

## Combined Degree Information (BSCE + MSCE)

A combined BSCE + MSCE program is available for outstanding Civil Engineering undergraduate students. This program can be completed in as little as five years (non-thesis and thesis option MSCE are possible) and result in receiving both the BSCE and MSCE degrees.

The BSCE + MSCE program is a mechanism for:

1. Providing a seamless transition from the BSCE to the MSCE program.
2. Stimulating interest in graduate study for advanced learning which would support more opportunities and faster advancement in professional and research/academic careers.
3. Allowing for special recognition of high levels of academic achievement.

The BSCE + MSCE program allows students to take up to 6 credits of graduate coursework (500 level) toward their BSCE professional elective requirement. These same 6 credits likewise count toward the MSCE degree. Additional graduate level courses taken in excess of the bachelor's requirement may also be eligible for transfer to a MSCE program for a total of 12 credits (6 dual counted; 6 excess).
Interested students typically apply as an "internal CE applicant" in the first half of their junior year with a cumulative undergraduate GPA of 3.3 or higher. If the internal application is accepted, the student would apply for combined program status to begin in the first semester of their senior year with a formal Graduate School application. A GPA of 3.3 must be maintained during combined program status. Grades of "B" or better must be earned for any course that would be dual counted.

Complete details of the combined BSCE + MSCE program can be found at https://engineering.purdue.edu/CE/Academics/Graduate/Combined-BSCE-MSCE. Questions about this information should be directed to cegrad@purdue.edu

## Pre-Requisite Information

For pre-requisite information, log in to mypurdue.purdue.edu and click here.

## World Language Courses

World Language proficiency requirements vary by program. The following list is inclusive of all world languages PWL offers for credit; for acceptable languages and proficiency levels, see your advisor. (ASL-American Sign Language; ARAB-Arabic; CHNS-Chinese; FR-French; GER-German; GREK-Greek(Ancient); HEBR-Hebrew(Biblical); HEBR-Hebrew(Modern); ITAL-Italian; JPNS-Japanese; KOR-Korean; LATN-Latin; PTGS-Portuguese; RUSS-Russian; SPAN-Spanish)

## Critical Course

The ♦ course is considered critical.

In alignment with the Degree Map Guidance for Indiana's Public Colleges and Universities, published by the Commission for Higher Education (pursuant to HEA 1348-2013), a Critical Course is identified as "one that a student must be able to pass to persist and succeed in a particular major. Students who want to be nurses, for example, should know that they are expected to be proficient in courses like biology in order to be successful. These would be identified by the institutions for each degree program."

## Disclaimer

The student is ultimately responsible for knowing and completing all degree requirements. Consultation with an advisor may result in an altered plan customized for an individual student. The myPurduePlan powered by DegreeWorks is the knowledge source for specific requirements and completion.

Comparative information about Purdue University and other U.S. educational institutions is also available through the College Navigator tool, provided by the National Center for Education Statistics, and through the U.S. Department of Education College Scorecard.